OPEN AGENT HACKATHON 2026 · REASONING ARCHITECTURE

Pulsar 5

A disaster response planning agent that **changes its mind correctly** when the world changes.

> Hurricane Ian, September 2022. Early forecasts pointed at Tampa Bay. Landfall came 100 miles south, at Fort Myers. This is a system built for that gap.

WHY THIS IS AN AGENT PROBLEM, NOT A FORECASTING PROBLEM

The forecast was not wrong. The plans built on it were stale.

> “the cone of uncertainty did contain the landfall location for **all advisory cycles**”
>
> — National Hurricane Center, post-storm report on Ian

The forecast was normal

Ian's 48-hour track error was **67.8 n mi** — about 78 miles, in line with the preceding five-year average. At 24 hours it beat the average.

The surge forecast tripled

At Fort Myers Beach it went **4–7 ft → 8–12 ft → 12–18 ft** in about a day as the track slid south. Measured peak: 12.70 ft.

30 miles decided everything

Pirate Harbor measured **4.47 ft**. Sanibel, under 30 miles away, **12.58 ft**. Tampa Bay sat **4.38 ft below** normal.

Every advisory carried the information needed to protect Lee County. Nobody was missing data — the plans simply **outlived the facts they were built on**. You cannot forecast your way out of that. You have to replan.

Source: NHC Tropical Cyclone Report AL092022, Forecast and Warning Critique · Tables 5a, 7

Architecture

six agents · one shared state · three engines

ZETARIS — the only data path

storm_advisories · shelters · hospitals · acs_vulnerability · staging_sites · road_events · disaster_declarations

WORLD MODEL — shared state every agent reads and writes

beliefs with provenance · decisions with dependencies · conflict log

Analyst
tool loop over Zetaris

Planner
authors every action

Simulator
plain code, no model

Critic
can reject the plan

Risk
the human gate

Belief Reviser
targeted replan

Planner ⇄ Simulator ⇄ Critic iterates, bounded at 3. H-MEM carries lessons across runs; Markovian keeps the hourly replay flat.

It is a loop, not a chain

from a real trace, run ian-landfall-minus-48h

ITERATION 1 — REJECTED

Planner opens 3 shelters — opening one costs staff and cots, so it starts lean.

Simulator 14,184 expected evacuees against 5,441 open spaces. Short by 8,743.

Critic refuses at confidence 0.56 and sends the shortfall back as an instruction.



ITERATION 2 — ACCEPTED

Planner raises the opening budget for exactly the reason given, and reaches 23,556 spaces.

Critic accepts at 0.78. **Risk** then holds 11 of 29 actions for a human and lets 18 commit.

The first pass is a genuine planning trade-off, not a staged failure. Opening every shelter immediately is not free, so the system starts conservative and lets the simulator prove it wrong.

The change of mind

POST /events — advisory 23 arrives

BEFORE · advisory 15, Tampa cone

Hillsborough 3 · Pinellas 3 · Sarasota 3 · Charlotte 2

Lee County: no action

AFTER · advisory 23, Fort Myers cone

Lee 3 · Charlotte 3 · Sarasota 2 · Pinellas 1

Hillsborough: stood down

27
beliefs changed

11
decisions invalidated

36
decisions preserved

+4 -6
actions added, removed

Sarasota was in both cones, so nothing touched it. Thirty-six decisions survived because no fact they cited actually changed. That is the difference between revising and starting over.

Reliability scoreboard

`35 perturbed scenarios · python scripts/eval.py`

0	0	3/3	129	1.76
constraint violations left unhandled	irreversible actions auto-committed	correct abstentions on missing data	plans the Critic caught before commit	mean iterations to an accepted plan

Abstention is a measured behaviour

Lose a **core** source — the advisory, the zones, the shelters — and the run declines to plan rather than inventing a number. Lose an **optional** one and it degrades: the plan is made, and the trace records what was missing.

34 tests, and a fresh-clone check

Engine conformance, loop behaviour, escalation rules. Cloned fresh with an empty `.env` and both services stopped, a full 29-action plan still comes back in 5 ms.

PULSAR 5

At advisory 15, with every forecast pointing at Tampa, it had already flagged Lee County to the operator.

```
distant_vulnerable_zone: Zone FL-071-0500 (Lee) scores 0.83 on social
vulnerability but lies 48 mi outside the current cone. No evacuation is
warranted yet; pre-positioning there is an operator call.
```

Not a prediction. The Critic simply refused to let a high-vulnerability area disappear just because the current cone missed it — and handed the judgement to the human, where it belongs.

github.com/lazizbekravshanov/pulsar5